

Integrable Systems — Exercise Sheet 8

Lax pairs, and the spectral parameter

Variant: Questions only

Conventions, as in Lecture 8: $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$, so that $\dot{f} = \{f, H\}$; a Lax pair satisfies $\dot{L} = [M, L] = ML - LM$ along the flow; and by Lecture 8's isospectrality theorem the $\text{tr } L^k$ are then first integrals, and $L(t) = U(t)L(0)U(t)^{-1}$ with $\dot{U} = MU$, $U(0) = \mathbb{1}$. For the Toda chain, Flaschka's variables are $a_n = e^{(q_n - q_{n+1})/2}$ and $b_n = p_n$.

Exercise 1 (the oscillator's Lax pair). Let $H = \frac{1}{2}(p^2 + q^2)$, so that $\dot{q} = p$ and $\dot{p} = -q$. (i) Lecture 8's first step: show that $L = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$ has $L^2 = 2H \mathbb{1}$. What are its eigenvalues? (ii) The second step: show that $\dot{L} = [M, L]$ is a linear system for the entries of M ; that if M solves it, so does $M + \alpha \mathbb{1} + \beta L$ for any functions α, β ; and that the only antisymmetric solution continuous on the phase space is $M = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. (iii) Show that e^{tM} is the rotation $R(t/2)$ through $t/2$, and that $R(\theta)L(p, q)R(\theta)^T = L(p', q')$, where (p', q') is the vector (p, q) turned through 2θ . Deduce the motion from $L(t) = U(t)L(0)U(t)^{-1}$.

Exercise 2 (the open Toda chain with three particles). Let $H = \frac{1}{2}(p_1^2 + p_2^2 + p_3^2) + e^{q_1 - q_2} + e^{q_2 - q_3}$. (i) Show that Hamilton's equations give $\dot{a}_n = \frac{1}{2}a_n(b_n - b_{n+1})$ and $\dot{b}_n = a_{n-1}^2 - a_n^2$, with $a_0 = a_3 = 0$. With Lecture 8's 3×3 matrix L (symmetric tridiagonal, diagonal b_n , off-diagonal a_n), take M antisymmetric and tridiagonal with unknown entries, $M_{12} = -m_1 = -M_{21}$ and $M_{23} = -m_2 = -M_{32}$, and derive m_1 and m_2 from $\dot{L} = [M, L]$, entry by entry. Which entries determine them, and which only test the result? (ii) Compute $\text{tr } L$, $\text{tr } L^2$ and $\text{tr } L^3$ in terms of q and p . Which of them did you know already? (iii) Use the Cayley–Hamilton theorem to write $\text{tr } L^4$ as a polynomial in $s_k = \text{tr } L^k$, $k = 1, 2, 3$.

Exercise 3 (a gauge with ones above the diagonal). Take the L and M of Exercise 2 and $D = \text{diag}(1, a_1, a_1 a_2)$. (i) Show that $\tilde{L} = DLD^{-1}$ has b_n on the diagonal, 1 above it and $a_n^2 = e^{q_n - q_{n+1}}$ below it. (ii) Compute $\tilde{M} = DMD^{-1} + \dot{D}D^{-1}$, which by Lecture 8's proposition on gauge transformations satisfies $\dot{\tilde{L}} = [\tilde{M}, \tilde{L}]$. (iii) Show that $\tilde{M} = \frac{1}{2}b_1 \mathbb{1} - \frac{1}{2}\tilde{L} + \tilde{L}_-$, where $\tilde{L}_-$ is the strictly lower triangular part of $\tilde{L}$, and deduce $\dot{\tilde{L}} = [\tilde{L}_-, \tilde{L}]$. Which first integrals survive the gauge transformation, and what is lost?

Exercise 4 (the Euler top in Lax form). Euler's equations for the free top are $\dot{\ell} = \ell \times \omega$ with $\omega_i = \ell_i/I_i$, $I_1, I_2, I_3 > 0$ (Lecture 8's reading). For $a \in \mathbb{R}^3$ let $\hat{a}$ be the antisymmetric matrix with $\hat{a}v = a \times v$. (i) Show that $[\hat{a}, \hat{b}] = \widehat{a \times b}$, and deduce that $L = \hat{\ell}$ satisfies $\dot{L} = [M, L]$ with $M = -\hat{\omega}$. (ii) Show that $\det(\eta - L) = \eta^3 + |\ell|^2 \eta$. What does the spectrum of L conserve? (iii) Show that $H = \sum_i \ell_i^2 / (2I_i)$ is conserved, and that unless $I_1 = I_2 = I_3$ it is not a function of $|\ell|^2$.

Exercise 5 (the ring, its site matrix, and zero curvature). (i) For the Toda ring with $N = 3$, $H = \frac{1}{2} \sum_n p_n^2 + e^{q_1 - q_2} + e^{q_2 - q_3} + e^{q_3 - q_1}$, show that $a_1 a_2 a_3 = 1$ with $a_3 = e^{(q_3 - q_1)/2}$. Write Lecture 8's L (with a_3 in the corners) and M (antisymmetric, $M_{n, n+1} = -\frac{1}{2}a_n$, indices modulo 3), and verify $\dot{L} = [M, L]$. (ii) Why is $N = 2$ degenerate? Compute $\text{tr } L^2$ for the $N = 2$ ring. (iii)

Show that the site matrix $L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}$ has determinant 1, that $\psi_n = e^{q_n/2} \phi_n$ turns the eigenvalue recursion $a_{n-1}\psi_{n-1} + b_n\psi_n + a_n\psi_{n+1} = \lambda\psi_n$ into $\phi_{n+1} = (\lambda - p_n)\phi_n - e^{q_{n-1}-q_n}\phi_{n-1}$, and that then $\Phi_{n+1} = L_n(\lambda)\Phi_n$ for $\Phi_n = (\phi_n, -e^{q_{n-1}}\phi_{n-1})^T$. (iv) With $V_n(\lambda) = \begin{pmatrix} -\lambda/2 & -e^{-q_n} \\ e^{q_{n-1}} & \lambda/2 \end{pmatrix}$, compute $V_{n+1}L_n - L_nV_n$, and show that $\dot{L}_n = V_{n+1}L_n - L_nV_n$ for all n and every λ exactly when the ring's equations of motion hold.

Exercise 6 (what a Lax pair does not give). (i) Show that every Hamiltonian has a Lax pair of size 1. What does it conserve? (ii) For a free particle in $\mathbb{R}^3$, $H = \frac{1}{2}|p|^2$, with $\ell = q \times p$, show that $L = \text{diag}(\ell_1, \ell_2, \ell_3)$, $M = 0$ is a Lax pair, and compute $\{\text{tr } L, \text{tr } L^3\}$ using $\{\ell_i, \ell_j\} = \sum_k \epsilon_{ijk} \ell_k$ (Sheet 2, Exercise 1). Are the spectral invariants of a Lax pair in involution in general? (iii) In three sentences, say what the spectral parameter λ of the site matrix buys, with the Toda ring as the example.

Exercise 7 ★. Manakov's spectral parameter. For the top of Exercise 4, let $A = \text{diag}(A_1, A_2, A_3)$ with $A_i = \frac{1}{2}(I_1 + I_2 + I_3) - I_i$, and $\Omega = \hat{\omega}$. (i) Show that $A_2 + A_3 = I_1$ (and cyclically) and that $\hat{\ell} = A\Omega + \Omega A$. (ii) Show that $L(\lambda) = \hat{\ell} + \lambda A^2$ and $M(\lambda) = -(\Omega + \lambda A)$ satisfy $\dot{L}(\lambda) = [M(\lambda), L(\lambda)]$ for every λ . (iii) Compute $\det(\eta - L(\lambda))$ as a polynomial in η and λ , and show that its coefficients determine H , for all $I_1, I_2, I_3 > 0$. (iv) For a symmetric top, $I_1 = I_2 \neq I_3$, show that ℓ_3 is conserved and that ℓ_3^2 is determined by the coefficients of $\det(\eta - L(\lambda))$.